



Unit 2: Experimentation and Measurement Essential Knowledge and Guided Notes

Read Textbook: (Link on top of the CANVAS Unit 2 Calendar) CK Chemistry Chapter 3:
Measurement: <https://flexbooks.ck12.org/cbook/ck-12-chemistry-flexbook-2.0/>

Watch Helpful Unit 2 Videos: Posted on CANVAS

Highlight Essential Knowledge: Use a different color marker or highlighter.

Experimental Design:

1. The **independent variable** is the variable that is changed
2. The **dependent variable** is the variable that is measured.
3. The **control** is the level (group) of the independent variable that others are compared to. It is often the "no treatment" group.
4. **Constants** are variables that are not allowed to change in order to see the effect of just the IV on the DV.
5. A **hypothesis** is a reasoned prediction of the outcome of an experiment
6. **Repeated trials** are done to verify the validity of the experimental method. A good experimental method will result in good repeatability.
7. Scientific data is organized in **data tables** and **graphs**.
8. Graphed data gives a picture of a relationship.
9. To graph data, the following standards apply
 - a. Independent variable (horizontal x-axis)
 - b. Dependent variable (vertical y-axis)
 - c. Scale and units are proportional
 - d. Regression lines (best-fit curve or line) are used to determine relationships between variables.

Scientific notation, SI units and temperature scales:

1. Measurements must be expressed in these **International System of Units (SI)** units.
2. Measurements of quantity include length (meter: m), volume (liter: L, dm³), mass (kilogram or gram: kg,g), temperature (Kelvin: K), time (second), and energy (Joule: J).
3. **Scientific notation** is used to write very small and very large numbers.
 - a. Multiplication of exponents is done by addition of the exponents
 - b. Division of exponents is done by subtraction of the exponent in the denominator from the exponent in the numerator.
4. The following measurements use the specified equipment
 - a. volume: graduated cylinder (L)
 - b. mass: electronic balances or scales (g)
 - c. temperature: thermometer and/or temperature probe (K)
5. The temperature scales used in science are **Kelvin (K) and Celsius (°C)**
 - a. Temperature is a measure of molecular motion (kinetic energy)
 - b. 0 K or -273°C is called **absolute zero**. At that temperature the kinetic energy of matter is also zero.
 - c. To convert between the two scales use this equation: **K = °C + 273**

Accuracy, precision, % error and types of error:

1. Mathematical procedures are used to validate data, including percent error to evaluate accuracy and demonstrate precision (reproducibility) in measurement.
 - a. **Accuracy** is closeness to the true value of a measurement.
 - b. **Precision** as reproducibility in measurement
 - c. To calculate percent error.

$$\% \text{ error} = \frac{(\text{Accepted value} - \text{Experimental value})}{\text{Accepted value}} \times 100$$

2. Errors in measurement may be random or systematic.
 - a. **Random errors** have an equal probability of being too high or too low. Almost all measurements have some degree of random error. Random error can be minimized with enough trials (take the average of the trials).
 - b. **Systematic errors** tend to either trend toward being too high or too low. Systematic errors do not average out with repeated trials. For example, a balance is not properly calibrated, so it may always systematically read too high.

Significant Figures:

1. Significant digits (figures) of a measurement are the number of known digits together with one estimated digit. The last digit of any valid measurement must be estimated and is therefore uncertain.
2. There are rules for determining how many significant figures are in a number:
 - a. **Non-zero digits are always significant.**
 - b. **Any zeros between two significant digits are significant. (sandwich zeros)**
 - c. **Leading zeros in front of a decimal are not significant**
 - d. **When there are final (trailing) zeros in a measurement the zeros are ONLY significant if the number contains a decimal point**
 - e. **Exact numbers such as 4 apples or definitions of units such as 1 g = 1000 mg have an unlimited or infinite number of significant figures.**
3. Follow these rules to perform calculations using significant figures:

Addition and Subtraction

When measured quantities are used in addition or subtraction, the uncertainty is determined by the absolute uncertainty in the least precise measurement (not by the number of significant figures). Sometimes this is considered to be the number of digits after the decimal point.

$$32.01 \text{ m} + 5.325 \text{ m} + 12 \text{ m}$$

Added together, you will get 49.335 m, but the sum should be reported as 49 meters.

Multiplication and Division

When experimental quantities are multiplied or divided, the number of significant figures in the result is the same as that in the quantity with the smallest number of significant figures.

If, for example, a speed calculation is made in which 25.624 m is divided by 25 sec, the speed should be reported as 1.0 m/sec, not as 1.000 m/sec or 1 m/sec.

Dimensional Analysis:

1. Dimensional Analysis – the method of converting units.
2. Conversion Factor – fractional quantity used to convert units.

•When we are given a quantity in some units, we can convert these to another unit:

Use a T chart: Fractional Conversion Factor

$$\text{Given Quantity} \times \frac{\text{Wanted Unit}}{\text{Given Unit}} = \text{Wanted Quantity}$$

Ex: How many inches is 3.00 centimeters? (Conversion: 1 inch = 2.54 cm)

$$3.00 \text{ cm} \times \frac{1 \text{ inch}}{2.54 \text{ cm}} = 1.18 \text{ in}$$

Density:

1. Density is defined as the mass per unit volume ($\frac{g}{mL}$ or $\frac{g}{cm^3}$)
2. $D = \frac{\text{mass}}{\text{volume}}$

Lesson #1 Guided Notes - Experimental Design Review

1. Define
 - a. Independent Variable:
 - b. Dependent Variable:
 - c. Constants:
 - d. Control:
2. Problem #1 - What is the effect of different concentrations of fertilizer on the height and health of plants?
 - a. Independent Variable:
 - b. Dependent Variable:
 - c. Constants:
 - d. Control:
3. Problem #2 - A student is studying the effects of several solutions on the prevention of the browning of apples. The student used solutions having different pH values and immersed three apple slices in equal volumes of each solution. He then timed to see how long it would take until the apples showed browning.
 - a. Independent Variable:
 - b. Dependent Variable:
 - c. Constants:
 - d. Control:
4. Define Hypothesis:
 - a. Format:
 - b. Write your hypothesis:
5. Define: Null Hypothesis:
 - a. Format:
 - b. Write your null hypothesis:
6. _____ are done to verify the validity of the experimental method. A good experimental method will result in good repeatability.
7. Scientific data is organized in _____ and _____.
Graphed data gives a picture of a relationship.

8. To graph data the following standards apply:
 - a. Independent variable (_____ axis)
 - b. Dependent variable (_____ axis)
 - c. Scale and units are _____
 - d. Regression lines (best-fit curve or line) are used to _____.

9. What is the meaning of DRY MIX?

10. Problem #3 - Sketch the graph shown. What are the IV and DV in the graph shown?

Lesson #2 Guided Notes - SI units, temperature scales and scientific notation

1. Both _____ and _____ must be included to express a quantity properly.
2. Identify the **number** and the **unit** in each quantity.
 - a. one dozen eggs
 - b. 2.54 centimeters
 - c. 1 box of pencils
 - d. 88 meters per second
3. _____ is a system of measurement based on the metric system.
4. What is a derived unit of measurement?
5. Be familiar with the base units and the SI prefixes:

INTERNATIONAL SYSTEM OF UNITS (SI)

SI Base Units

Base Quantity	Name	Symbol
Length	meter	m
Mass	kilogram	kg
Time	second	s
Electric current	ampere	A
Thermodynamic temperature	kelvin	K
Amount of substance	mole	mol
Luminous intensity	candela	cd

SI Derived Units

Derived Quantity	Name	Symbol	Equivalent SI Units
Frequency	hertz	Hz	s ⁻¹
Force	newton	N	m•kg•s ⁻²
Pressure	pascal	Pa	N/m ²
Energy	joule	J	N•m
Power	watt	W	J/s
Electric charge	coulomb	C	s•A
Electric potential	volt	V	W/A
Electric resistance	ohm	Ω	V/A
Celsius temperature	degree Celsius	°C	K*

*Unit degree Celsius is equal in magnitude to unit kelvin

SI Prefixes

Factor	Name	Symbol	Numerical Value
10 ¹²	tera	T	1 000 000 000 000
10 ⁹	giga	G	1 000 000 000
10 ⁶	mega	M	1 000 000
10 ³	kilo	k	1 000
10 ²	hecto	h	100
10 ¹	deka	da	10
10 ⁻¹	deci	d	0.1
10 ⁻²	centi	c	0.01
10 ⁻³	milli	m	0.001
10 ⁻⁶	micro	μ	0.000 001
10 ⁻⁹	nano	n	0.000 000 001
10 ⁻¹²	pico	p	0.000 000 000 001

Adapted from NIST Special Publication 811.
SI rules and style conventions recommend using spaces rather than commas to separate groups of three digits.

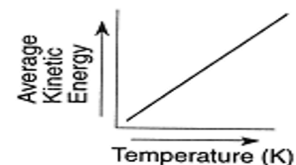
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6. The temperature scales used in science are _____ and _____. Temperature is a measure of molecular motion called _____.

7. There is a _____ relationship between temperature and kinetic energy.



8. 0 K or -273°C is called _____. At that temperature, the kinetic energy of matter is also zero.

9. To convert between the two scales use this equation: $K = \text{_____} + \text{_____}$

10. Try these conversions:

20°C = -10°C = 300K = 100°C = 150K =

11. Why do we need scientific notation?

12. Scientific notation is a method of representing very large or very small numbers

$$M \times 10^n$$

13. What does M represent?

14. What does n represent?

15. Convert to scientific notation:

a. 2,500,000,000 =

b. 0.0000579 =

16. Multiplication Rule - _____ Exponents

a. $10^3 \times 10^4 =$

b. $10^5 \times 10^2 \times 10^{-3} =$

17. Division Rule - _____ Exponents

a. $10^5 \div 10^3 =$

b. $10^7 \div 10^{12} =$

c. $10^8 \div 10^{-3} =$

d. $(10^6 \times 10^4) \div (10^3 \times 10^2) =$

18. Try these (No calculator)

a. $(2 \times 10^3) \times (4 \times 10^5) =$

b. $(4 \times 10^3) \times (3 \times 10^2) =$

c. $(4 \times 10^5) \div (2 \times 10^7) =$

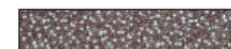
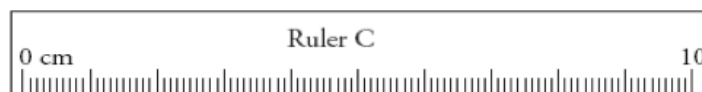
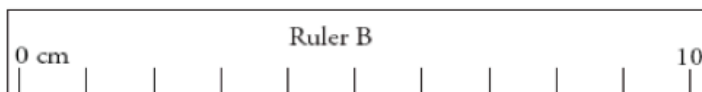
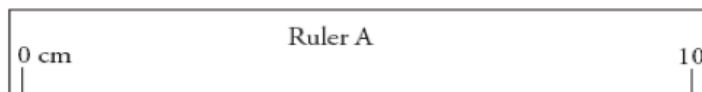
Lesson #3 Guided Notes - Accuracy, precision, % error and types of error

1. **Accuracy** is a measure of how close a measurement comes to the _____.
2. **Precision** is a measure of how close a series of measurements _____.
3. Can you have a single measurement that is accurate? _____
4. Can you have a single measurement that is precise? _____
5. Problem #1 - Sam weighs a mass of known weight of 50.0 grams on an analytical scale that he has not calibrated recently. He weighs the mass 5 times and gets the following readings: 150 gram, 151 grams, 149 grams, 150 grams, 150 grams. What can you say about Sam's scale? _____
6. How do you evaluate accuracy and precision?
 - a. To evaluate the **accuracy** of a measurement, the measured value must be compared to the _____.
 - b. To evaluate the **precision** of a measurement, you must compare the values of _____.
7. The _____ (or theoretical) value is the correct value based on reliable references.
8. The _____ value is the value measured in the lab.
9. The difference between the experimental value and the accepted value is called the _____
 - a. To calculate, take the absolute value of the difference between the two values.
 - b. $Error = |experimental\ value - accepted\ value|$
10. What is the equation for the percent error?
11. Problem #2 - A scale that needs calibration reads a standard weight of 150 gram as 155 grams. What is the scale's error? _____ What is the scales % error?
12. Errors in measurement may be random or systematic.
 - a. **Random Errors** have an equal probability of being too high or too low. Almost all measurements have some degree. Random error can be minimized with enough trials (take the average of the trials). An example would be _____
 - b. **Systematic Errors** tend to either tend toward being too high or too low. Systematic error does not average out with repeated trials. An example would be _____

Lesson #4 Guided Notes - Counting significant figures

Examine what taking a valid measurement from this Model 4:

Model 4 – Valid Measurements



Valid Measurements	Invalid Measurements
3 cm	2.5 cm
2 cm	3.00 cm
	$3\frac{1}{4}$ cm
	3.33 cm
3.2 cm	3 cm
3.1 cm	3.25 cm
3.3 cm	3.20 cm
3.21 cm	3.2 cm
3.22 cm	3.215 cm
3.20 cm	3.205 cm

-
- According to this model
 - In a valid measurement, you should record _____ estimated digit.
 - In a valid measurement, the estimated digit is the _____ digit.
 - In a valid measurement, the estimated digit is _____ of the smallest mark.
- Significant digits (or figures) in a measurement are all of the marked (certain) digits that are read directly and the next estimated (uncertain) digit. Using this definition, write the length of the plank correctly using significant digits. _____ cm



4. Rules for Counting Significant Figures:

Significant digits, which are also called significant figures, are very important in Chemistry. Each recorded measurement has a certain number of significant digits Any digit that is actually measured or estimated will be considered significant. Placeholders, or digits that have not been measured or estimated, are not considered significant. The rules for counting significant digits are:

- Non-zero digits are always significant.**
- Any zeros between two significant digits are significant. (sandwich zeros)**
- Leading zeros in front of a decimal are not significant**
- When there are final (trailing) zeros in a measurement the zeros are ONLY significant if the number contains a decimal point**
- Exact numbers such as 4 apples or definitions of units such as $1\text{ g} = 1000\text{ mg}$ have an unlimited or infinite number of significant figures.**

Examples

1.33 g = 3 significant figures
1.023 L = 4 significant figures
21.00 kg = 4 significant figures
0.021 kg = 2 significant figures
0.0210 kg = 3 significant zeros
2100 kg = 2 significant figures
2100. kg = 4 significant figures

5. Practice Questions:

5.0201g has _____ significant figures

0.011 seconds has _____ significant figures

0.0350 mg has _____ significant figures

5000 kg has _____ significant figures

5 apples has _____ significant figures

Lesson#5 Guided Notes - Significant figures and Calculations

Addition and Subtraction: The number of decimal places in the result equals the number of decimal places in the least accurate measurement.

6.8g + 11.934 g = 18.734 or 18.7g (tenths place)

<u>Calculation</u>	<u>Calculator says:</u>	
3.24 m + 7.0 m	10.24 m	→
100.0 g - 23.73 g	76.27 g	→
0.02 cm + 2.371 cm	2.391 cm	→
713.1 L - 3.872 L	709.228 L	→
1818.2 lb + 3.37 lb	1821.57 lb	→
2.030 mL - 1.870 mL	0.16 mL	→

Multiplication and Division: # sig figs in the result equals the number in the least number of significant figures in the measurements used in the calculation.

$$6.38 \text{ cm} \times 2.0 \text{ cm} = 12.76 = 13 \text{ cm}^2 \text{ (2 sig figs)}$$

<u>Calculation</u>	<u>Calculator says:</u>	
3.24 m x 7.0 m	22.68 m ²	→
100.0 g ÷ 23.7 cm ³	4.219409283 g/cm ³	→
0.02 cm x 2.371 cm	0.04742 cm ²	→
7.10 m ÷ 3.0 s	2.366666667 m/s	→
1818.2 lb x 3.23 ft	5872.786 lb·ft	→
1.030 g ÷ 2.87 mL	0.358850174 g/mL	→

Lesson #6 Guided Notes - Dimensional Analysis

1. Definitions

a. Dimensional Analysis – _____

b. Conversion Factor – _____

2. Write two ways of writing the conversion factor: **1 inch = 2.54 cm**

3. Example "T-Chart"

a. Convert 12 inches = _____ cm

b. Convert 12 cm = _____ inches

4. Example multi-step "T -Chart" : How many inches in 145 meters?

Conversion Factors: 1 inch = 2.54 cm, 100 cm = 1 meter

5. Try this one with a T-chart: You decide to make your career wrapping hamburgers. Every 3 hours you wrap 55 hamburgers. You work 8 hours a day, 5 days a week. You get paid \$455 dollars every two weeks after taxes. How many burgers will you have to wrap to make your first \$1,000,000?

Conversion factors: (Convert \$1,000,000 dollars into burgers!)

3 hours = ____ hamburgers

8 hours = ____ day

5 days = ____ week

\$455 = ____ weeks

Lesson #7 Guided Notes - Density

1. Definition of Density:
2. Why is density a derived unit of measurement?
3. What are the units of density? Solid = Liquid = $1 \text{ cm}^3 = 1 \text{ mL}$
4. What is the equation for solving for density?
5. Draw the Density Triangle and then solve below:
 - a. $D =$
 - b. $V =$
 - c. $m =$
6. Example Problems
 - a. If the volume is 2.0 ml and its mass is 4.6 grams what is the density of the substance?
 - b. If the density is 4.0 g/mL and the volume is 1.5 mL what is the mass?
 - c. If the mass is 4.5 g and the density is 1.5 g/mL what is the volume?
7. How do you determine the density of a liquid? What is a meniscus?
8. How do you determine the density of an irregularly shaped solid?
9. Example: The initial volume of a graduated cylinder is 11.5 mL. A solid object is dropped in and the water level rises to 14.0 mL. What is the volume of the solid object? If the same object's mass is 10.00 grams what is the density of the solid? Round to the correct number of significant figures. Volume = _____ Density = _____